

Assignment-2

Linear and Nonlinear Regression

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Python 3.13.3

Code:

```
import numpy as np
import matplotlib.pyplot as plt

#defining functions for each problem
def solve_problem_1():
    """Concrete strength vs Cement content (Linear)"""
    print("\n--- Problem 1: Concrete Strength Prediction ---")
    C = np.array([300, 320, 340, 360, 400])
    fc = np.array([28.0, 30.0, 33.0, 35.0, 40.0])

    # Linear regression:  $fc = a_0 + a_1 \cdot C$ 
    coeffs = np.polyfit(C, fc, 1)
    a1, a0 = coeffs

    prediction = a0 + a1 * 380
    print(f"Model:  $fc = \{a_0:.4f\} + \{a_1:.4f\} \cdot C$ ")
    print(f"Predicted fc for C=380: {prediction:.2f} MPa")
    plot_results(C, fc, coeffs, "Cement Content (kg/m3)", "Strength (MPa)", "Problem 1")

def solve_problem_2():
    """Young's Modulus (Linear Stress-Strain)"""
    print("\n--- Problem 2: Young's Modulus Estimation ---")
    strain = np.array([0.25, 0.50, 0.75, 1.00, 1.25]) * 1e-3
    stress = np.array([50, 100, 150, 205, 250])

    # Linear regression:  $stress = a + b \cdot strain$ 
    coeffs = np.polyfit(strain, stress, 1)
    b, a = coeffs

    print(f"Estimated Young's Modulus (E): {b/1000:.2f} GPa")
    plot_results(strain, stress, coeffs, "Strain", "Stress (MPa)", "Problem 2")

def solve_problem_3():
```

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"""Beam Deflection (Linear) and Stiffness"""
print("\n--- Problem 3: Beam Deflection and Stiffness ---")
P = np.array([0, 10, 20, 30, 40])
delta = np.array([0.05, 1.02, 2.01, 3.05, 4.02])

coeffs = np.polyfit(P, delta, 1)
a1, a0 = coeffs
stiffness = 1/a1
print(f"Model: delta = {a0:.4f} + {a1:.4f}*P")
print(f"Estimated Stiffness (k = P/delta): {stiffness:.2f} kN/mm")
plot_results(P, delta, coeffs, "Load (kN)", "Deflection (mm)", "Problem 3")

```

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def solve_problem_4():
    """Concrete strength vs Cement content (Linear)"""
    print("\n--- Problem 4: Concrete Strength Prediction ---")
    C = np.array([300, 320, 340, 360, 400])
    fc = np.array([28.0, 30.0, 33.0, 35.0, 40.0])

    # Linear regression: fc = a0 + a1*C
    coeffs = np.polyfit(C, fc, 1)
    a1, a0 = coeffs

    prediction = a0 + a1 * 380
    print(f"Model: fc = {a0:.4f} + {a1:.4f}*C")
    print(f"Predicted fc for C=380: {prediction:.2f} MPa")
    plot_results(C, fc, coeffs, "Cement Content (kg/m3)", "Strength (MPa)", "Problem 4")

```

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def solve_problem_5():
    """Multivariate Linear Regression for Concrete Strength"""
    print("\n--- Problem 5: Multivariate Linear Regression ---")
    C = np.array([300, 320, 340, 360, 380, 400])
    wc = np.array([0.55, 0.53, 0.50, 0.48, 0.46, 0.44])
    fc = np.array([20.5, 22.5, 26.0, 28.2, 31.5, 34.1])

    A = np.column_stack([np.ones(len(C)), C, wc])
    coeffs, _, _, _ = np.linalg.lstsq(A, fc, rcond=None)
    a0, a1, a2 = coeffs

    pred_fc = a0 + a1*350 + a2*0.49
    print(f"Model: fc = {a0:.4f} + {a1:.4f}*C + {a2:.4f}*(w/c)")
    print(f"Predicted fc for C=350, w/c=0.49: {pred_fc:.2f} MPa")

```

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def solve_problem_6():
    """Quadratic Regression: Strength vs Age"""

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print("\n--- Problem 6: Quadratic Regression ---")
t = np.array([7, 10, 14, 21, 28])
fc = np.array([24, 28, 32, 38, 42])

coeffs = np.polyfit(t, fc, 2)
a2, a1, a0 = coeffs

pred_18 = a2*(18**2) + a1*18 + a0
print(f"Model: fc = {a0:.4f} + {a1:.4f}*t + {a2:.4f}*t^2")
print(f"Predicted fc at 18 days: {pred_18:.2f} MPa")
plot_results(t, fc, coeffs, "Age (days)", "Strength (MPa)", "Problem 6", poly_order=2)

def solve_logarithmic_problem(t, y, label_x, label_y, title, pred_t):
    """Helper for Problems 7, 8, 9, 11 (Logarithmic)"""
    ln_t = np.log(t)
    coeffs = np.polyfit(ln_t, y, 1)
    a1, a0 = coeffs
    prediction = a0 + a1 * np.log(pred_t)
    print(f"Model: {label_y} = {a0:.4f} + {a1:.4f}*ln(t)")
    print(f"Predicted value at t={pred_t}: {prediction:.2f}")

def solve_problem_7():
    print("\n--- Problem 7: Logarithmic Strength Growth ---")
    t = np.array([7, 10, 14, 21, 28])
    fc = np.array([24.0, 27.0, 30.0, 35.0, 38.0])
    solve_logarithmic_problem(t, fc, "Age (days)", "Strength (MPa)", "Problem 7", 18)

def solve_problem_8():
    print("\n--- Problem 8: Logarithmic Creep Strain ---")
    t = np.array([1, 3, 7, 14, 30])
    eps = np.array([120, 180, 240, 300, 360])
    solve_logarithmic_problem(t, eps, "Time (days)", "Creep Strain", "Problem 8", 10)

def solve_problem_9():
    print("\n--- Problem 9: Logarithmic Settlement ---")
    t = np.array([1, 2, 5, 10, 20])
    s = np.array([4.0, 5.2, 6.8, 8.0, 9.1])
    solve_logarithmic_problem(t, s, "Time (days)", "Settlement (mm)", "Problem 9", 15)

def solve_exponential_problem(t, y, label_x, label_y, title, pred_t):
    """Helper for Problems 10, 12 (Exponential)"""
    ln_y = np.log(y)
    coeffs = np.polyfit(t, ln_y, 1)
    B, lnA = coeffs
    A = np.exp(lnA)

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prediction = A * np.exp(B * pred_t)
print(f"Model: {label_y} = {A:.4f} * e^{(B:.4f)*t}")
print(f"Predicted value at t={pred_t}: {prediction:.2f}")

```

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def solve_problem_10():
    print("\n--- Problem 10: Exponential Strength Model ---")
    t = np.array([7, 10, 14, 21, 28])
    fc = np.array([20.0, 23.0, 27.0, 33.0, 38.0])
    solve_exponential_problem(t, fc, "Age (days)", "Strength (MPa)", "Problem 10", 18)

```

```

def solve_problem_11():
    print("\n--- Problem 11: Logarithmic Traffic Growth ---")
    t = np.array([1, 2, 4, 6, 10])
    adt = np.array([12000, 13700, 15500, 16500, 17800])
    solve_logarithmic_problem(t, adt, "Years", "ADT (veh/day)", "Problem 11", 8)

```

```

def solve_problem_12():
    print("\n--- Problem 12: Exponential Decay Amplitude ---")
    t = np.array([0, 1, 2, 3, 4])
    amp = np.array([10.0, 7.4, 5.5, 4.1, 3.0])
    solve_exponential_problem(t, amp, "Time (s)", "Amplitude (mm)", "Problem 12", 5)

```

```

def plot_results(x, y, coeffs, xlabel, ylabel, title, poly_order=1):
    plt.scatter(x, y, color='red', label='Data Points')
    p = np.poly1d(coeffs)
    x_line = np.linspace(min(x), max(x), 100)
    plt.plot(x_line, p(x_line), label='Regression Line')
    plt.xlabel(xlabel)
    plt.ylabel(ylabel)
    plt.title(title)
    plt.legend()
    plt.show()

```

#asking user to choose a problem to be solved

```

def main():
    problems = {
        "1": solve_problem_1, "2": solve_problem_2, "3": solve_problem_3,
        "4": solve_problem_4, "5": solve_problem_5, "6": solve_problem_6,
        "7": solve_problem_7, "8": solve_problem_8, "9": solve_problem_9,
        "10": solve_problem_10, "11": solve_problem_11, "12": solve_problem_12
    }

```

```

while True:
    print("\n--- Problem Set - 02 ---")
    print("Enter Problem Number (1-12) or 'q' to quit.")
    choice = input("Choice: ").strip()

```

```

if choice.lower() == 'q':
    break
elif choice in problems:
    problems[choice]()
else:
    print("Invalid choice.")

if __name__ == "__main__":
    main()

```

Problems and Outputs

Problem:1 At 28 days, concrete cubes were tested for mixes with different cement contents. Assume a linear relationship (valid over the tested range):

$$f_c = a_0 + a_1 C$$

where C is cement content (kg/m^3) and f_c is compressive strength (MPa).

Task: Using least squares, estimate a_0 and a_1 . Then predict f_c for $C = 380 \text{ kg/m}^3$.

Mix No.	Cement C (kg/m^3)	Strength f_c (MPa)
1	300	28.0
2	320	30.0
3	340	33.0
4	360	35.0
5	400	40.0

--- Problem Set - 02 ---

Enter Problem Number (1-12)

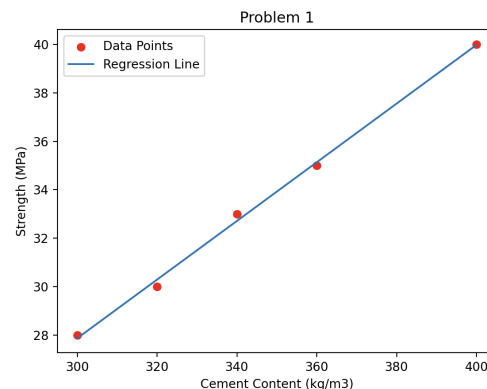
Choice: 1

|

--- Problem 1: Concrete Strength Prediction ---

Model: $f_c = -8.4054 + 0.1209 \cdot C$

Predicted f_c for $C=380$: 37.55 MPa



Problem:2 A steel specimen is tested in tension. The following stress–strain data are recorded in the elastic range.

Test No.	Strain ε ($\times 10^{-3}$)	Stress σ (MPa)
1	0.25	50
2	0.50	100
3	0.75	150
4	1.00	205
5	1.25	250

Estimate Young's modulus E using linear regression. Assume a linear relationship between stress and strain:

$$\sigma = a + b\varepsilon$$

where:

- σ = stress (MPa)
- ε = strain
- b = slope of regression line $\approx E$

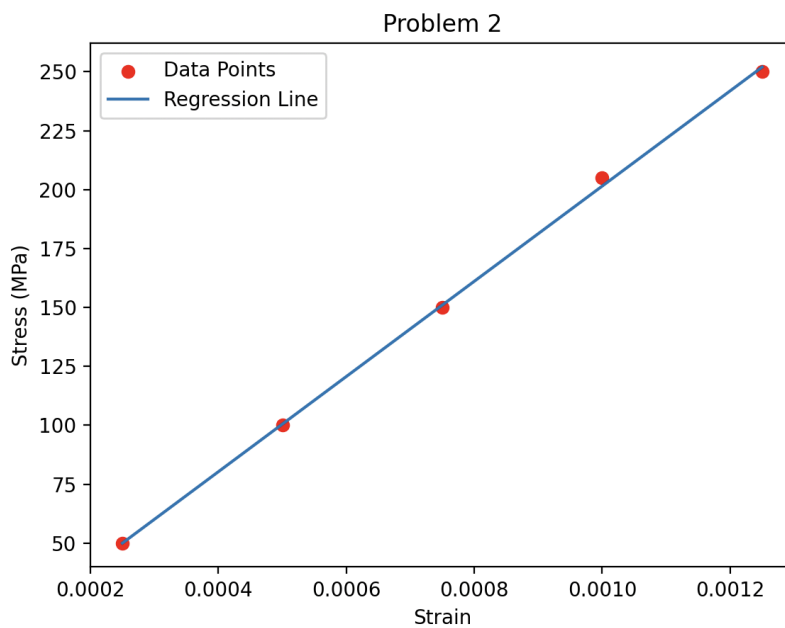
--- Problem Set - 02 ---

Enter Problem Number (1-12)

Choice: 2

--- Problem 2: Young's Modulus Estimation ---

Estimated Young's Modulus (E): 202.00 GPa



Problem:3 A simply supported steel beam is tested under gradually increasing load. Mid-span deflection is measured for each applied load.

Assume a linear model in the service (elastic) range:

$$\delta = a_0 + a_1 P$$

where P is load (kN) and δ is deflection (mm).

Task: Using least squares, estimate a_0 and a_1 , and interpret stiffness.

Test No.	Load P (kN)	Deflection δ (mm)
1	0	0.05
2	10	1.02
3	20	2.01
4	30	3.05
5	40	4.02

--- Problem Set - 02 ---

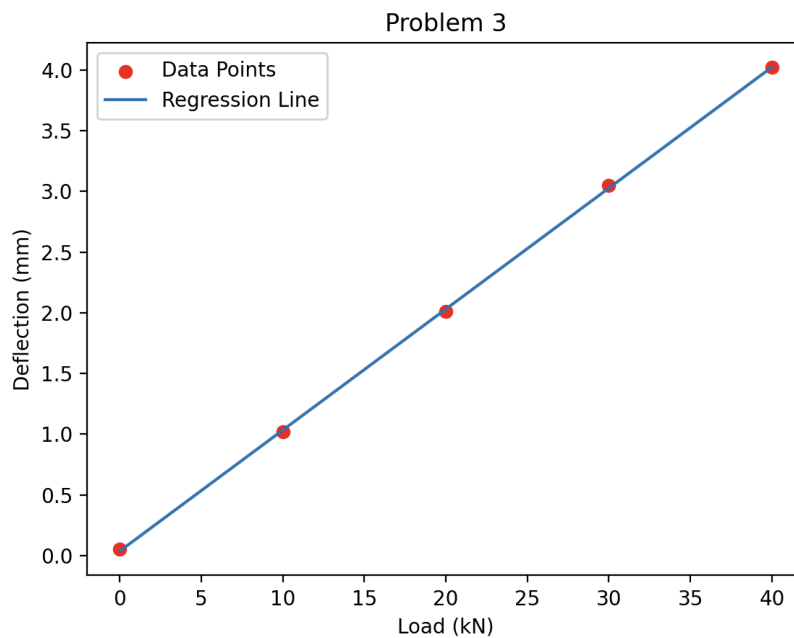
Enter Problem Number (1-12)

Choice: 3

--- Problem 3: Beam Deflection and Stiffness ---

Model: $\delta = 0.0360 + 0.0997 \cdot P$

Estimated Stiffness ($k = P/\delta$): 10.03 kN/mm



Problem:4 — Same as Problem1

Problem:5 At 28 days, estimate compressive strength f_c (MPa) using:

- $x_1 = C$ = cement content (kg/m³)
- $x_2 = w/c$ = water-cement ratio

Assume a multivariate linear model:

$$f_c = a_0 + a_1C + a_2(w/c)$$

Data (6 mixes):

Mix	C (kg/m ³)	w/c	f_c (MPa)
1	300	0.55	20.5
2	320	0.53	22.5
3	340	0.50	26.0
4	360	0.48	28.2
5	380	0.46	31.5
6	400	0.44	34.1

Task: Find a_0, a_1, a_2 using least squares and predict f_c for $C = 350$, $w/c = 0.49$.

--- Problem Set - 02 ---

Enter Problem Number (1-12)

Choice: 5

--- Problem 5: Multivariate Linear Regression ---

Model: $fc = 11.8591 + 0.0968*C + -37.7273*(w/c)$

Predicted fc for $C=350$, $w/c=0.49$: 27.26 MPa

Problem:6 Concrete cube compressive strength increases with curing age, but the trend shows curvature. Fit a quadratic regression model:

$$f_c(t) = a_0 + a_1t + a_2t^2$$

where:

- t = age (days)
- f_c = cube strength (MPa)

Task:

- Estimate a_0, a_1, a_2 by least squares.
- Predict f_c at $t = 18$ days.

Test No.	Age t (days)	Strength f_c (MPa)
1	7	24
2	10	28
3	14	32
4	21	38
5	28	42

--- Problem Set - 02 ---

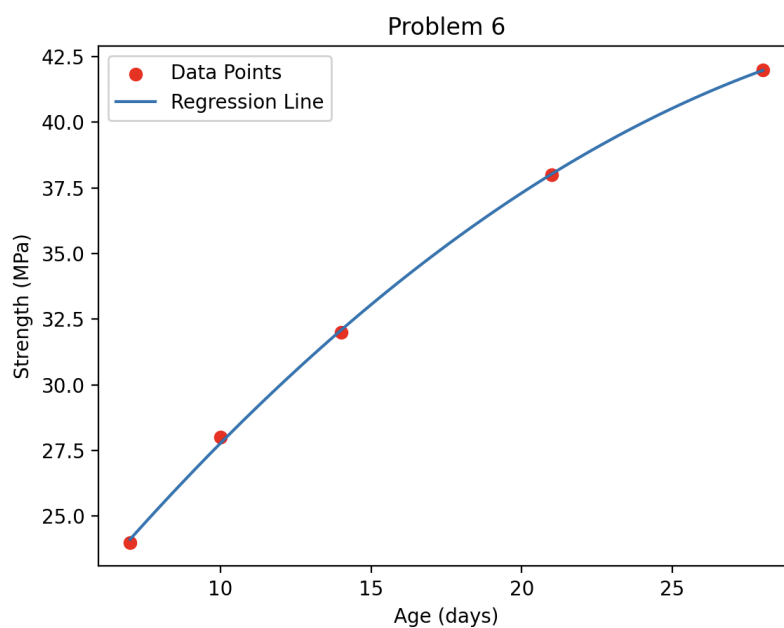
Enter Problem Number (1-12)

Choice: 6

--- Problem 6: Quadratic Regression ---

Model: $fc = 14.1450 + 1.5687*t + -0.0205*t^2$

Predicted fc at 18 days: 35.73 MPa



Problem:7 Early-age concrete strength often increases rapidly and then slows down.
Fit a logarithmic regression model:

$$f_c(t) = a_0 + a_1 \ln(t)$$

where:

- t = curing age (days), with $t > 0$
- f_c = cube compressive strength (MPa)

Task: Estimate a_0, a_1 using following data and predict f_c at $t = 18$ days.

Test No.	Age t (days)	Strength f_c (MPa)
1	7	24.0
2	10	27.0
3	14	30.0
4	21	35.0
5	28	38.0

--- Problem Set - 02 ---

Enter Problem Number (1-12)

Choice: 7

--- Problem 7: Logarithmic Strength Growth ---

Model: Strength (MPa) = 3.6071 + 10.2504*ln(t)

Predicted value at t=18: 33.23

Problem:8 Concrete creep strain often increases rapidly at early times and then slows down. Assume a logarithmic model for creep strain:

$$\varepsilon_c(t) = a_0 + a_1 \ln(t)$$

where:

- t = time after loading (days), $t > 0$
- ε_c = creep strain (microstrain)

Task: Estimate a_0, a_1 using following data and predict ε_c at $t = 10$ days

Test No.	t (days)	ε_c (microstrain)
1	1	120
2	3	180
3	7	240
4	14	300
5	30	360

--- Problem Set - 02 ---

Enter Problem Number (1-12)

Choice: 8

--- Problem 8: Logarithmic Creep Strain ---

Model: Creep Strain = 110.4331 + 71.3099*ln(t)

Predicted value at t=10: 274.63

Problem:9 A plate load / settlement observation indicates that settlement increases rapidly at early time and then slows down. Assume a logarithmic model:

$$s(t) = a_0 + a_1 \ln(t)$$

where:

- t = time (days), $t > 0$
- s = settlement (mm)

Data:

Observation	t (days)	s (mm)
1	1	4.0
2	2	5.2
3	5	6.8
4	10	8.0
5	20	9.1

Task: Estimate a_0, a_1 and predict settlement at $t = 15$ days.

--- Problem Set - 02 ---

Enter Problem Number (1-12)

Choice: 9

--- Problem 9: Logarithmic Settlement ---

Model: Settlement (mm) = 4.0185 + 1.7113*ln(t)

Predicted value at t=15: 8.65

Problem:10 Concrete strength may be approximated over a limited age range by an exponential model:

$$f_c(t) = Ae^{Bt}$$

where:

- t = curing age (days)
- f_c = cube compressive strength (MPa)

Task: Estimate A and B by least squares using following data and predict f_c at $t = 18$ days.

Test No.	t (days)	f_c (MPa)
1	7	20.0
2	10	23.0
3	14	27.0
4	21	33.0
5	28	38.0

--- Problem Set - 02 ---

Enter Problem Number (1-12)

Choice: 10

--- Problem 10: Exponential Strength Model ---

Model: Strength (MPa) = 16.9203 * $e^{(0.0302*t)}$

Predicted value at $t=18$: 29.16

Problem:11 In a corridor, measured Average Daily Traffic (ADT) increases rapidly in the initial years due to network effects, and later the growth slows down. Assume a logarithmic model:

$$\text{ADT}(t) = a_0 + a_1 \ln(t)$$

where:

- t = years since opening ($t > 0$)
- ADT = average daily traffic (vehicles/day)

Data:

Year since opening t	1	2	4	6	10
ADT (veh/day)	12000	13700	15500	16500	17800

Task: Estimate a_0, a_1 and predict ADT at $t = 8$ years.

--- Problem Set - 02 ---

Enter Problem Number (1-12)

Choice: 11

--- Problem 11: Logarithmic Traffic Growth ---

Model: $\text{ADT (veh/day)} = 11982.7357 + 2524.5969 \cdot \ln(t)$

Predicted value at $t=8$: 17232.49

Problem:12 In a free-vibration test of a structural system, the peak displacement amplitude decays with time due to damping. Assume an exponential decay model:

$$A(t) = A_0 e^{-\lambda t}$$

Equivalently, in regression form:

$$y = Ae^{Bt} \quad \text{where} \quad A = A_0, \quad B = -\lambda$$

Data (measured peak amplitude):

Observation	Time t (s)	Amplitude y (mm)
1	0	10.0
2	1	7.4
3	2	5.5
4	3	4.1
5	4	3.0

Task: Estimate A and B using least squares and predict amplitude at $t = 5$ s.

--- Problem Set - 02 ---

Enter Problem Number (1-12)

Choice: 12

--- Problem 12: Exponential Decay Amplitude ---

Model: Amplitude (mm) = 10.0079 * e^(-0.2998*t)

Predicted value at t=5: 2.23